

accept waits for the
full 1024-sample pack

Legacy

Trigger

Accept gate
builder idle

1024-sample builder
capture + pack now

Output FIFO +
Hermes-input drain

busy_count

samples already buffered;
accept now and pack later

full_count

2k ring + 50% overlap

Trigger

Accept gate
256-sample spacing + safety

2048-sample
ring RAM

Trigger queue
4 deep

Deferred 512-sample
packer

Same output FIFO +
Hermes-input drain

spacing_count
ring_count

queue_count

output_count
full_count